

Conformal prediction as a solution to distribution shifts in medical image analysis

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Abstract

Before medical AI systems can be widely deployed in clinics, there is a need for uncertainty quantification and improved handling of distribution shifts. Conformal prediction (CP) is an uncertainty quantification method that is highly useful but requires datasets to exhibit exchangeability, something that is broken by distribution shifts. In this work, we employ CP on real-world pathology image data to investigate distribution shifts’ impact on the validity of CP. We find that the main driver for loss of exchangeability is disagreements between pathologists, rather than medical differences between patients.

Keywords: Artificial intelligence, digital pathology, conformal prediction, medical imaging

1. Introduction

Automated medical systems using artificial intelligence (AI) are becoming wide-spread in research and beginning to enter clinical practice [Fahim et al. \(2025\)](#); [Rajpurkar et al. \(2022\)](#); [Andersen et al. \(2025\)](#). Despite this, several barriers are known that should be overcome before wide-spread use in high-stakes clinical situations becomes tenable [Hassan et al. \(2024\)](#), something which is echoed in the EU AI act, citing the need for transparency, human oversight, and robustness [European Parliament and the Council of the European Union \(2024\)](#). Necessary for shattering these barriers is reliable uncertainty quantification [Banerji et al. \(2023\)](#) and a proper handling of distribution shifts [Jeong et al. \(2024\)](#).

Distribution shift in machine learning refers to changes in the joint distribution of input data and labels between the environment in which a machine is produced and the environment in which it is deployed. It can be specified as domain shift, label shift, and concept shift. Respectively, they refer to changes in the underlying distribution of input data, changes in the underlying distribution of labels, and changes in the underlying conditional distribution of labels given the input data or vice-versa [Moreno-Torres et al. \(2012\)](#).

Conformal prediction (CP) is an uncertainty quantification method in which an allowed degree of uncertainty is specified by the user and mathematically ensured by the CP algorithm [Vovk et al. \(2022\)](#). It replaces point predictions with prediction intervals, and always produces valid prediction intervals as long as the calibration data and subsequent data on which the predictor is deployed have the statistical property of exchangeability. Generally, exchangeability is broken by distribution shifts [Barber et al. \(2023\)](#).

In this work, we used CP on a prostate cancer Gleason grading classifier using real-world pathology image data (whole slide images), assessed the effect of different types of distribution shift on exchangeability, and identified medical or practical factors associated with such shifts.

2. Methods

We calibrated a conformal predictor on a machine learning model for prostate cancer Gleason grading using 7,290 (initially 14,580 split 50/50 between training and testing) whole slide images from the Stockholm3 dataset [Grönberg et al. \(2015\)](#). Data from this cohort was used in training the classifier but the CP data was held-out during training, and the entire dataset was graded by the same expert pathologist. The conformal predictor was then tested on several cohorts including Stockholm3 (data held out during calibration), cohorts that were in-distribution for the classifier but out-of-distribution for the conformal predictor, and cohorts that were out-of-distribution for both the classifier and the conformal predictor. One external cohort consisting of rare prostate cancer subtypes was graded by the same pathologist as Stockholm3, while all others were graded by other pathologists. However, a sample of cases from a variety of cohorts were re-graded by the Stockholm3 pathologist.

We observed coverage violation in the different cohorts using Mondrian (class-conditional) CP, which prevents loss of exchangeability due to label shift. All experiments use a pre-specified error rate of 5%.

3. Results

Testing the conformal predictor on Stockholm3 data ($n = 7,290$), which is exchangeable with the calibration data by design, yielded a coverage of 95.5%. This dropped to 92.8% when considering data from cohorts that was used during model training but not CP calibration ($n = 1,714$), and further to 88.9% when testing on data fully external to both the classifier and the conformal predictor ($n = 6,296$). The coverage on the cohort of rare subtypes ($n = 371$) was 69.5%. Finally, the coverage on the sample that was re-graded by the Stockholm3 pathologist ($n = 348$) was 83.3% when using the original labels and 93.4% when using the re-graded labels.

4. Discussion

The coverage drop observed on data external to calibration and/or model development indicates that distribution shifts did break the exchangeability assumption required for validity of conformal predictors. Generally, these data subsets were not exchangeable with the calibration data due to both domain shift and concept shift. Notably, however, in the final experiment, the increase in coverage when applying the re-graded labels versus the original labels was not due to a difference in domain shift, since the same data were observed in the two tests. Hence, the source of loss of exchangeability was concept shift induced by pathologist disagreements or mislabeling. Furthermore, the coverage was nearly at the target of 95% with the re-graded labels, implying that pathologist disagreements or mislabeling may be the most important factor both for loss of exchangeability as well as for model performance during testing of machine learning classification models in computational pathology. We note that in that case, this is something which impacts model evaluation heavily, while not constituting a significant problem in model deployment, where predictive models are no longer trying to match the operating pathologist but merely predict the patient’s medical state.

References

- Adam Andersen, Ruiping Huang, and Edward Jiusi Liu. The Application of Artificial Intelligence in Healthcare Practice: An Umbrella Review, October 2025. URL <https://medrxiv.org/lookup/doi/10.1101/2025.10.05.25337356>.
- Christopher R. S. Banerji, Tapabrata Chakraborti, Chris Harbron, and Ben D. MacArthur. Clinical AI tools must convey predictive uncertainty for each individual patient. *Nature Medicine*, 29(12):2996–2998, December 2023. ISSN 1078-8956, 1546-170X. doi: 10.1038/s41591-023-02562-7. URL <https://www.nature.com/articles/s41591-023-02562-7>.
- Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2), April 2023. ISSN 0090-5364. doi: 10.1214/23-AOS2276. URL <https://projecteuclid.org/journals/annals-of-statistics/volume-51/issue-2/Conformal-prediction-beyond-exchangeability/10.1214/23-AOS2276.full>.
- European Parliament and the Council of the European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act) (Text with EEA relevance), June 2024. URL <http://data.europa.eu/eli/reg/2024/1689/oj>. Legislative Body: CONSIL, EP.
- Yosri A. Fahim, Ibrahim W. Hasani, Samer Kabba, and Waleed Mahmoud Ragab. Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives. *European Journal of Medical Research*, 30(1):848, September 2025. ISSN 2047-783X. doi: 10.1186/s40001-025-03196-w. URL <https://eurjmedres.biomedcentral.com/articles/10.1186/s40001-025-03196-w>.
- Henrik Grönberg, Jan Adolfsson, Markus Aly, Tobias Nordström, Peter Wiklund, Yvonne Brandberg, James Thompson, Fredrik Wiklund, Johan Lindberg, Mark Clements, Lars Egevad, and Martin Eklund. Prostate cancer screening in men aged 50–69 years (STHLM3): a prospective population-based diagnostic study. *The Lancet Oncology*, 16(16):1667–1676, December 2015. ISSN 14702045. doi: 10.1016/S1470-2045(15)00361-7. URL <https://linkinghub.elsevier.com/retrieve/pii/S1470204515003617>.
- Masooma Hassan, Andre Kushniruk, and Elizabeth Borycki. Barriers to and Facilitators of Artificial Intelligence Adoption in Health Care: Scoping Review. *JMIR Human Factors*, 11:e48633, August 2024. ISSN 2292-9495. doi: 10.2196/48633. URL <https://humanfactors.jmir.org/2024/1/e48633>.
- Hyewon Jeong, Sarah Jabbour, Yuzhe Yang, Rahul Thapta, Hussein Mozannar, William Jongwon Han, Nikita Mehandru, Michael Wornow, Vladislav Lialin, Xin Liu, Alejandro Lozano, Jiacheng Zhu, Rafal Dariusz Kocielnik, Keith Harrigian, Hao-ran Zhang, Edward Lee, Milos Vukadinovic, Aparna Balagopalan, Vincent Jeanselme, Katherine Matton, Ilker Demirel, Jason Fries, Parisa Rashidi, Brett Beaulieu-Jones, Xuhai Orson Xu, Matthew McDermott, Tristan Naumann, Monica Agrawal, Marinka Zitnik, Berk Ustun, Edward Choi, Kristen Yeom, Gamze GURSOY, Marzyeh Ghassemi, Emma Pierson, George Chen, Sanjat Kanjilal, Michael Oberst, Linying Zhang, Harvineet Singh,

Tom Hartvigsen, Helen Zhou, and Chinasa T. Okolo. Recent Advances, Applications, and Open Challenges in Machine Learning for Health: Reflections from Research Roundtables at ML4H 2023 Symposium, 2024. URL <https://arxiv.org/abs/2403.01628>. Version Number: 2.

Jose G. Moreno-Torres, Troy Raeder, Rocío Alaiz-Rodríguez, Nitesh V. Chawla, and Francisco Herrera. A unifying view on dataset shift in classification. *Pattern Recognition*, 45(1):521–530, January 2012. ISSN 00313203. doi: 10.1016/j.patcog.2011.06.019. URL <https://linkinghub.elsevier.com/retrieve/pii/S0031320311002901>.

Pranav Rajpurkar, Emma Chen, Oishi Banerjee, and Eric J. Topol. AI in health and medicine. *Nature Medicine*, 28(1):31–38, January 2022. ISSN 1078-8956, 1546-170X. doi: 10.1038/s41591-021-01614-0. URL <https://www.nature.com/articles/s41591-021-01614-0>.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic learning in a random world*. Springer, Cham, second edition edition, 2022. ISBN 978-3-031-06651-1 978-3-031-06648-1.